

ELENA PAGNIN

Assistant Professor at Chalmers
University of Technology

Personal Info

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No. publications: **30+**

No. citations: **480+**

h-index: **13**

Education & Titles

since 2022	Assistant Professor Tenure-Track (Forskarassistent)	Chalmers (SE)
2020 - 2022	Associate Senior Lecturer (Biträdande Universitetslektor)	Lund University (SE)
2019 - 2020	Post-Doctoral Researcher in Cryptography	Aarhus University (DK)
Feb - Mar		
2014 - 2019	Ph.D. in Computer Science (Cryptography)	Chalmers (SE)
May - Jan	Thesis Title: " <i>Be more and be merry: enhancing data and user authentication in collaborative settings</i> " 	
	Supervisor: A. Sabelfeld	
2016	Licentiate Degree of Engineering	Chalmers (SE)
Aug 25th	Thesis Title: " <i>Authentication under Constraints</i> " 	
	Supervisor: A. Mitrokotsa	
2012 - 2013	Project Officer (paid Researcher Assistant position)	Nanyang TU (SG)
Aug - Feb	Master thesis project under Prof. F. Oggier supervision	

Recent Achievements (Grants & Awards)

- 2025 The Cryptography course (TDA352/DIT352) appears in the recommended list of educational resources of **campus totalforsvar** (strategic collaboration between Swedish educational institutions aimed at strengthening Sweden's total defense through education and research)
- 2023-2024 Parental Leave (10 months)
- 2023 **Co-PI** in a seed funding awarded by Chalmers' Areas of Advance on the topic *Towards a Multi-Layer Security Vision for Transportation Systems in the 6G Era* **(0.5M SEK)**
- 2023-2027 **PI** of the prestigious Starting Grant awarded by the Swedish Research Council (**VR starting**) on *Progressive Verification of Cryptographic Schemes* **(4M SEK)**
- 2020-2022 **PI** in a personal grant awarded to strategic research areas by the Excellence Center **ELLIIT** **(1.6M SEK)**
- 2020 **Best Paper Award** for the full-version of "*Multi-Key Homomorphic Authenticators*" (extended abstract appeared in ASIACRYPT16) granted by the IET Information Security Journal. [This kind of Premium Award is given to recognize the best research papers published during the last two years]

Supervision of PhD Students (= I am the main advisor, = I co-supervise together with)

- 2024 - now **Lucia Lavagnino**: *Transparency Logs* () Chalmers (SE)
Planned PhD Defense: Nov 2029
- 2023 - now **Adrian Perez Keilty**: *Progressive Verification of Cryptographic Schemes* () Chalmers (SE)
Planned PhD Defense: 20 Aug 2028
- 2023 - now **Hanna Ek**: *Advanced Properties for Digital Signatures* () Chalmers (SE)
Planned PhD Defense: 31 Dec 2027
- 2019 - now **Ivan Oleynikov**: *Privacy-Preserving Location Proximity Testing* () Chalmers (SE)
Licentiate: 30 Sept 2022. Planned PhD Defense: December 2025
- 2022 - now **Arthur Nijdam**: *Secure Machine Learning for the Medical Sector* ( P. Stankovski-Wagner) Lund University (SE)
Planned PhD Defense: September 2027

Graduated Students

- 2025.03.07 **Joakim Brorsson**: *On the Trustworthiness of Trusted Third Parties* ( P. Stankovski-Wagner) Lund University (SE)
- 2023.03.21 **Martin Gunnarsson**: *Securing IoT Systems* ( Prof. C. Gehrman) Lund University (SE)
- 2023.02.28 **Hadi Sehat**: *Dual Deduplication in multi-client setting and its applications* ( Prof. D. Lucani) Aarhus University (DK)

Selection of Professional Activities

- 09.06.2025 Member of the examination committee for the PhD Defence of Joel Gärtner (upcoming) KTH (SE)
- 31.01.2024 Contributed to the **referral** to a new National law proposal on a Swedish electronic identification system *En säker och tillgänglig statlig e-legitimation* (SOU 2023:61)
- since 2021 **Program Committee Member** for: LatinCrypt25, ACNS25, LatinCrypt23, ACNS23, SEC@SAC22, ACISP21.
- since 2022 **Mentor** in the **WISE-WWACQT** mentorship program
- 11.10.2023 **Seminar** speaker at ICT Area of Advance full-day seminar on Navigating the Cybersecurity Landscape on *Fortifying the Digital Fortress: Provably Secure Cryptography* Chalmers (SE)
- 2023 **Invited Speaker** at **AI NORDIC POWWOW** on "Cybersecurity and AI for Company Security" Lund (SE)

2023	Seminar speaker at the BARC center on <i>Progressive Verification for Cryptographic Schemes</i>	KU Copenhagen (DK)
2021-2022	Scientific leader at the EIT Department for the SSF "SMARTY" grant RIT17-0035 (22M SEK)	Lund University (SE)
2022	Invited Speaker at the seminar series at <i>Protocol Labs: Extended Threshold Ring Signatures</i> 	(remote)
2022	Invited Speaker at the CRC seminar series at <i>TII Research Centers: Progressive and Efficient Verification</i> 	(remote)
2022	Invited Speaker at a National hearing of researchers on "Innovative Processes for Data Integrity and Data Sharing" organized by the Swedish Ministry for Integrity Protection (Integritetsskyddsmyndigheten - IMY)	Stockholm (SE)
2022	Invited Panelist on "Allyship and Inclusion" at CRYPTO22 	Santa Barbara (CA, USA)
2021	Invited Speaker at ELLIIT initiative on <i>Future-Oriented Research: "Enhancing Data Authentication"</i> 	Lund (SE)
2021	Participant in the <i>Researchers' Grand Prix: Security and Privacy in the Digital Era</i> 	Helsinborg (SE)
2020	Invited Speaker at <i>Framtidsveckan: AI, digitalisering och integritet – vad får vi för vår hälsodata?</i> Presentation and Panel Discussion on "Contact tracing apps and their security dilemmas" 	Lund (SE)
2019	Examiner for the PhD Defence of <i>Ijlal Loutfi</i> on "Trusted Execution on Commodity Devices"	University of Oslo (NO)

Teaching Experience		
2015-now	Master's Theses: Supervisor of 9, Examiner for 1.	Chalmers, Università di Milano, Lund University
2022-now	Course Responsible for the Masters level course <i>Cryptography</i> (TDA352/DIT352, 7.5hp)	Chalmers (SE)
2022	Lecturer and Co-instructor for the Masters level course <i>Advanced Cryptography</i> (EITN85, 7.5hp)	Lund University (SE)
2020 - 2022	Lecturer and Course Responsible for the Masters course <i>Advanced Web Security</i> (EITN41, 7.5hp)	Lund University (SE)
2021	Course Responsible for the PhD level course <i>Frontiers in Security Research</i> (EIT190F, 7.5hp) 	Lund University (SE)

Selected List of Publications	
AC24 	David, Engelmann, Frederiksen, Kohlweiss, Pagnin, and Volkhov: " <i>Updatable Privacy-Preserving Blueprints</i> ". In: <i>Annual International Conference on Theory and Application of Cryptology and Information Security - ASIACRYPT</i> (2024).
SCN24 	Baum, David, Elena Pagnin , and Takahashi: " <i>CaSCaDE:(Time-Based) Cryptography from Space Communications Delay</i> ". In: <i>International Conference on Security and Cryptography for Networks</i> (2024).
EuroS&P24 	Nelson, Elena Pagnin , and Askarov: " <i>Metadata Privacy Beyond Tunneling for Instant Messaging</i> ". In: <i>IEEE European Symposium on Security and Privacy - EuroS&P</i> (2024).
CT-RSA23 	Brorsson, David, Gentile, Elena Pagnin , and Stankovski-Wagner: " <i>PAPR: Publicly Auditable Privacy Revocation for Anonymous Credentials</i> ". In: <i>The Cryptographers' Track at RSA Conference</i> (2023).
ACNS22  	Boschini, Fiore, and Elena Pagnin : " <i>Progressive and Efficient Verification for Digital Signatures</i> ". In: <i>International Conference on Applied Cryptography and Network Security - ACNS</i> (2022).
CLOUD22 	Vestergaard, Elena Pagnin , Kundu, and Lucani: " <i>Secure Cloud Storage with Joint Deduplication and Erasure Protection</i> ". In: <i>International Conference on Cloud Computing - IEEE CLOUD</i> (2022).
PKC22  	Aranha, Hall-Andersen, Nitulescu, Elena Pagnin , and Yakoubov: " <i>Count Me In! Extendability for Threshold Ring Signatures</i> ". In: <i>International Conference on Practice and Theory of Public-Key Cryptography - PKC</i> (2022).
SCN20  	Lucani, Nielsen, Orlandi, Elena Pagnin , and Vestergaard: " <i>Secure Generalized Deduplication via Multi-Key Revealing Encryption</i> ". In: <i>International Conference on Security and Cryptography for Networks</i> (2020).
ESORICS20  	Oleynikov, Elena Pagnin , and Sabelfeld: " <i>Where Are You Bob? Privacy-Preserving Proximity Testing with a Napping Party</i> ". In: <i>European Symposium on Research in Computer Security – ESORICS</i> (2020).
AC16  	Fiore, Mitrokotsa, Nizzardo, and Elena Pagnin : " <i>Multi-key Homomorphic Authenticators</i> ". In: <i>Annual International Conference on Theory and Application of Cryptology and Information Security - ASIACRYPT</i> (2016).